

# Identification of the order distribution of linear systems by line-spectrum estimation on logarithmic spirals

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## Abstract

A linear time-invariant system whose transfer function admits the distributed-order representation  $H(s) = \int_{\mathbb{R}} \mathcal{A}(\alpha) s^\alpha d\alpha$  is characterized by its *order distribution*  $\mathcal{A}$ : integer-order systems correspond to Dirac combs supported on  $\mathbb{Z}$  whose weights are Laurent–Taylor coefficients, while fractional-order systems place mass at non-integer  $\alpha$ s. Because the order integral is a Laplace transform of  $\mathcal{A}$  in the variable  $\log s$ , sampling  $H$  on the unit circle  $s = e^{iu}$  and applying an FFT suggests itself as a direct, closed-form estimator of  $\mathcal{A}$  from measured data. We show that this natural estimator is subject to two structural obstructions: (i) a truncation bias, computed here in closed form, which corrupts every coefficient whenever the measured response has a non-zero steady state; and (ii) an aliasing theorem stating that any transform computed from time-domain data via  $s = e^{iu}$  is exactly  $2\pi$ -periodic, so its inverse Fourier transform is supported on  $\mathbb{Z}$  and non-integer orders can *never* appear as spectral lines, only as slowly decaying integer combs. We then repair both defects: subtracting the measured steady state reduces the integer-order problem to classical Cauchy-integral coefficient extraction (Lyness–Moler), and replacing the circle by a *logarithmic spiral*  $s = e^{(\sigma+i)u}$ ,  $\sigma \neq 0$ , breaks the periodicity, turning order recovery into a line-spectrum estimation problem solvable by matrix-pencil methods. Numerical experiments recover the seven leading Taylor coefficients of a second-order test system from its sampled step response with error  $\sim 10^{-8}$ , and recover the order pair  $(-1, -\frac{1}{2})$  (one integer, one fractional line) of a signal, from raw time samples with error  $\sim 2 \times 10^{-9}$  (noise-free) and  $\sim 5 \times 10^{-5}$  under  $10^{-4}$  relative measurement noise. As a calibration benchmark, the estimator recovers the complex dimensions of the Cantor fractal string—known in closed form—from geometric counting data alone, with damping estimates matching the exact value  $D = \log 2 / \log 3$  to six decimals, and, from spectral counting data, recovers the same dimensions together with channel gains matching independently computed values of the Riemann zeta function to four decimals. An open-source implementation is provided.

**Keywords:** fractional calculus, distributed-order systems, order distribution, system identification, Mellin transform, matrix pencil, Cauchy integrals, fractal strings, complex dimensions.  
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## 1 Introduction

Fractional-order models—systems governed by derivatives  $D^\alpha$  of non-integer order  $\alpha$ —arise in viscoelasticity, electrochemistry, anomalous diffusion, and control [8, 18]. A strictly more general class allows a *continuum* of orders: distributed-order operators  $\int \mathcal{A}(\alpha) D^\alpha y d\alpha$ , introduced by Caputo [5, 6, 7] and developed by Bagley and Torvik [2, 3] and by Lorenzo and Hartley

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[23]. In the Laplace domain (zero initial conditions) such a system has transfer function

$$H(s) = \int_{\mathbb{R}} \mathcal{A}(\alpha) s^\alpha d\alpha, \quad (1)$$

and the density  $\mathcal{A}$ , called the *order distribution* [13, 14], is the natural object of identification: an integer-order rational system corresponds to  $\mathcal{A}$  being a finite or geometric-weight Dirac comb on the integers, a commensurate fractional system to a comb on  $q\mathbb{Z}$  for some  $q \in (0, 1)$ , and genuinely distributed systems to continuous densities.

Identifying  $\mathcal{A}$  from measured data is a recognized problem. Published approaches discretize the order axis and fit the weights by least squares to frequency-response data [14, 26, 9], estimate discrete fractional orders by nonlinear optimization or instrumental variables [27, 28], read asymptotic Bode slopes, or, in the PDE setting, exploit analyticity of solutions in inverse-problem frameworks [22]. What makes a *closed-form* alternative tempting is a structural identity already noted in [13]: the right-hand side of (1) is a (bilateral) Laplace transform of  $\mathcal{A}$  with transform variable  $-\log s$ . Consequently, on the unit circle  $s = e^{iu}$  the identity reads  $H(e^{iu}) = \int \mathcal{A}(\alpha) e^{iu\alpha} d\alpha$ : a Fourier transform. One is led to the estimator

$$\mathcal{A}(\alpha) = \frac{1}{2\pi} \int_{\mathbb{R}} H(e^{iu}) e^{-iu\alpha} du, \quad (2)$$

computable by an FFT from samples of  $H$  on the circle, which in turn can be obtained from a measured step or impulse response by numerically evaluating its Laplace transform at  $s = e^{iu_k}$ . Orders present in the system would then appear as spectral lines of  $\mathcal{A}$ , integer or not.

This paper analyzes that program and establishes exactly how much of it survives contact with measured data. Our contributions are:

1. **A corrected integer-order pipeline with an exact bias formula.** The naive estimator applied to a step response with steady state  $y_\infty \neq 0$  suffers a truncation bias  $-y_\infty(-A)^{m+1}/(m+1)!$  at integer order  $m$ , where  $A$  is the measurement window (Proposition 3.2); the bias grows exponentially in  $A$  across the bins (like  $e^A$  up to algebraic factors) and dwarfs the true coefficients. Subtracting the measured steady state before transforming removes it entirely and reduces the method to Cauchy-integral coefficient extraction in the sense of Lyness and Moler [25] (Section 3).
2. **An aliasing obstruction.** Any function of  $u$  computed from time samples through  $s = e^{iu}$  is exactly  $2\pi$ -periodic; therefore its inverse Fourier transform is supported on  $\mathbb{Z}$  (Theorem 3.4). Non-integer orders are structurally invisible to the unit-circle readout of black-box data: a branch cut manifests itself only as an integer comb with slow  $\mathcal{O}(1/m)$  decay, for which we give a closed-form example. This clarifies which claims of the closed-form program hold for transfer functions known analytically (where the Riemann surface of  $s^\alpha$  can be unwound) but fail for data.
3. **A logarithmic-spiral estimator for fractional orders.** Sampling the data transform on  $s = e^{(\sigma+i)u}$  with  $\sigma \neq 0$  makes  $\log s$  single-valued and injective along the contour; a finite order comb then produces an exponential sum in the sample index, and the orders and weights follow from a matrix-pencil line-spectrum estimate [17] (Section 4). The estimator works directly on raw time samples, with the contour restricted to an arc in the right half-plane to keep the data transform convergent.
4. **Numerical validation and open code.** On the step response of  $H(s) = 1/(s^2+4s+5)$ , the corrected circle pipeline recovers  $a_0, \dots, a_6$  with maximum error  $1.2 \times 10^{-8}$ , the steady state being estimated from the data itself; the spiral estimator recovers the order pair  $(-1, -\frac{1}{2})$  of  $y(t) = 1/\sqrt{\pi t} + 2$  from time samples with error  $1.8 \times 10^{-9}$ , degrading to  $4.7 \times 10^{-5}$  under  $10^{-4}$  relative noise (Section 5).

5. **A calibration benchmark from fractal geometry.** The estimator is validated against complex dimensions of self-similar fractal strings, a setting where the “uniform damping” question—conjectural for the Riemann zeta function—is a *theorem*: for the Cantor string, the estimator recovers eight complex dimensions from geometric data with all damping estimates equal to the exact value  $D = \log 2 / \log 3$  to six decimals, and, from spectral counting data alone, recovers the same dimensions together with channel gains matching independently computed values of  $|\zeta|$  to four decimals in every mode—an acoustic measurement of the zeta function realizing the classical spectral factorization  $\zeta_\nu = \zeta_{\mathcal{L}} \cdot \zeta$  as a live transfer function (Section 5.5). A further experiment exhibits the estimator’s channel gain vanishing, as theory predicts, at a zeta zero.

Section 6 positions these results against the distributed-order identification literature and against the numerical analysis of Cauchy-integral coefficient extraction; proofs of the two propositions are collected in Appendix A.

## 2 The order-distribution representation

Throughout,  $s^\alpha = e^{\alpha \log s}$  with a branch of  $\log$  fixed by the contour under discussion, and  $\mathcal{A}$  is a tempered distribution on  $\mathbb{R}$ ; (1) is read in the distributional sense.

**Definition 2.1.** Let  $H$  be analytic on an open annulus  $\Omega = \{r_1 < |s| < r_2\}$ ,  $0 \leq r_1 < r_2 \leq \infty$ . A tempered distribution  $\mathcal{A}$  on  $\mathbb{R}$  is an *order distribution for  $H$  on  $\Omega$*  if

$$H(e^w) = \langle \mathcal{A}(\alpha), e^{\alpha w} \rangle$$

for all  $w$  with  $\log r_1 < \operatorname{Re} w < \log r_2$ .

**Example 2.2.** If  $H$  is analytic on an annulus containing  $|s| = 1$  and single-valued, its Laurent expansion  $H(s) = \sum_{n \in \mathbb{Z}} a_n s^n$  converges there with geometrically decaying  $a_n$ , and

$$\mathcal{A} = \sum_{n \in \mathbb{Z}} a_n \delta(\alpha - n)$$

is an order distribution for  $H$ : the classical case. For instance,  $H(s) = s^2 + k$  gives  $\delta(\alpha - 2) + k \delta(\alpha)$ ;  $H(s) = (s+1)/s$  gives  $\delta(\alpha) + \delta(\alpha+1)$ ;  $H(s) = 1/(s^2+2)$  gives  $\sum_{n \geq 0} (-1)^n 2^{-(n+1)} \delta(\alpha - 2n)$  on  $|s| < \sqrt{2}$ . A Caputo-type fractional term contributes mass at its order:  $H(s) = s^{1/2}$  has  $\mathcal{A} = \delta(\alpha - \frac{1}{2})$  on the universal cover of  $\mathbb{C} \setminus \{0\}$ .

*Remark 2.3* (Non-uniqueness). The order distribution depends on the annulus. E.g.  $1/(s^2+2)$  also equals  $\sum_{n \geq 0} (-1)^n 2^n s^{-2n-2}$  on  $|s| > \sqrt{2}$ , a different comb. Statements such as “the positions of the deltas reveal the orders of the system” are therefore well-posed only once a contour (equivalently, a strip of the Mellin variable) is fixed. In what follows the annulus is always the one containing the sampling contour.

Writing  $s = e^{iu}$ ,  $u \in \mathbb{R}$ , Definition 2.1 gives the Fourier pair

$$H(e^{iu}) = \int_{\mathbb{R}} \mathcal{A}(\alpha) e^{i\alpha u} d\alpha, \quad \mathcal{A}(\alpha) = \frac{1}{2\pi} \int_{\mathbb{R}} H(e^{iu}) e^{-i\alpha u} du, \quad (3)$$

where for multivalued  $H$  the left-hand side denotes the lift along the universal cover (i.e.  $u$  is *not* reduced modulo  $2\pi$ ). When  $\mathcal{A} = \sum_n a_n \delta(\alpha - n)$ , the weights are given by the Cauchy formula: substituting  $s = e^{iu}$ ,  $ds = is du$ ,

$$a_n = \frac{1}{2\pi i} \oint_{|s|=1} \frac{H(s)}{s^{n+1}} ds, \quad (4)$$

with the kernel  $s^{-\alpha-1}$  (not  $s^{-\alpha}$ ): the Jacobian  $ds/s$  is what aligns the continuous inversion with the classical coefficient formula. Under  $s = e^{-p}$ , (1) is a bilateral Laplace transform of  $\mathcal{A}$ ; the representation and its inversion are thus an instance of the Mellin correspondence [24], and the interpolation of power-series coefficients to continuous order is the territory of Ramanujan’s master theorem. None of this section is new mathematics; it fixes notation and conventions for the data-driven question that follows.

### 3 Recovery from measured data on the unit circle

Let the system be excited by a unit step,  $X(s) = 1/s$ , and let  $y(t)$ ,  $t \in [0, A]$ , be the measured response, so that  $Y(s) = H(s)/s$ . The data provide the *truncated* transform

$$\hat{Y}_A(s) = \int_0^A y(\tau) e^{-s\tau} d\tau, \quad (5)$$

an entire function of  $s$  computable by quadrature at any  $s_k = e^{iu_k}$ . The naive pipeline evaluates (5) on the circle, multiplies by  $s$  to undo the input, and applies a DFT across  $u$  to read off the comb weights. Two elementary identities explain its behavior exactly.

**Proposition 3.1** (Aliased-moment identity). *Let  $u_k = 2\pi k/N$ ,  $k = 0, \dots, N-1$ , and let  $\hat{y}(u) = \hat{Y}_A(e^{iu})$ . Then the  $m$ -th DFT bin of  $\{\hat{y}(u_k)\}$  equals*

$$\frac{1}{N} \sum_{k=0}^{N-1} \hat{y}(u_k) e^{-2\pi i m k/N} = \sum_{\substack{j \geq 0 \\ j \equiv m \pmod{N}}} \frac{(-1)^j}{j!} \int_0^A \tau^j y(\tau) d\tau.$$

*In particular the bins are (aliased, signed, truncated) time moments of the data, indexed by non-negative integers only.*

Time moments are the classical carriers of the  $s \rightarrow 0$  expansion of a transfer function in moment-matching identification and model reduction [12]; Proposition 3.1 shows the circle pipeline is an FFT reorganization of them. It also shows the pipeline returns nothing at negative orders—so the  $1/s$  of the step input cannot appear at  $\alpha = -1$  and must be handled separately—and nothing at non-integer orders, foreshadowing Theorem 3.4.

**Proposition 3.2** (Truncation bias). *Suppose  $y(t) = y_\infty + g(t)$  with  $|g(t)| \leq C e^{-\beta t}$ ,  $\beta > 0$  (stable step response,  $y_\infty = H(0)$ ). Then*

$$\hat{Y}_A(s) = \frac{y_\infty}{s} + G(s) - \frac{y_\infty e^{-sA}}{s}, \quad G(s) = \int_0^\infty g(\tau) e^{-s\tau} d\tau,$$

*and the entire function  $-y_\infty e^{-sA}/s + y_\infty/s = -y_\infty \sum_{m \geq 0} \frac{(-A)^{m+1}}{(m+1)!} s^m$  contributes the bias*

$$b_m = -y_\infty \frac{(-A)^{m+1}}{(m+1)!} \quad (6)$$

*to the coefficient of  $s^m$  for every  $m \geq 0$ —hence, up to the  $m + \ell N$  aliasing of Proposition 3.1 (negligible for  $N \gg A$ ), to DFT bin  $m$ —while the  $g$ -tail term is bounded by  $C e^{-(\beta + \cos u)A}/(\beta + \cos u)$  for  $\cos u > -\beta$ .*

For any useful window the bias (6) is catastrophic: with  $y_\infty = 0.2$  and  $A = 12$ ,  $b_0 = 2.4$  against a true coefficient  $a_1 = -0.16$  (the naive pipeline’s  $s$ -multiplication shifts  $b_m$  up one bin), and  $b_5 = -829.4$ . The experiment in Section 5 reproduces (6) to all printed digits. The repair is equally explicit:

**Corollary 3.3** (Corrected pipeline). *Estimate  $y_\infty$  from the tail of the record, form  $g = y - y_\infty$ , and*

$$H(e^{iu_k}) \approx y_\infty + e^{iu_k} \widehat{G}_A(e^{iu_k}), \quad \widehat{G}_A(s) = \int_0^A g(\tau) e^{-s\tau} d\tau, \quad (7)$$

*before the DFT. If every pole of  $H$  satisfies  $\text{Re}(\text{pole}) < -1$  (more generally, if the circle radius  $r$  is chosen with  $r < \beta$ ), the total error in each recovered coefficient is  $\mathcal{O}(e^{-(\beta-1)A})$  plus quadrature error, and the procedure coincides with Cauchy-integral coefficient extraction [25, 16, 10] applied to  $H$  itself, which the data reconstruct as  $y_\infty + s\widehat{G}_A(s)$ : here  $\widehat{G}_A(s)$  approximates the transient transform  $G(s) = (H(s) - H(0))/s$ , which is analytic at  $s = 0$ , with error bounded by the  $g$ -tail term of Proposition 3.2.*

The radius condition matters: for slow systems ( $\beta \leq 1$ ) the truncated transform of  $g$  no longer converges to  $G$  as  $A$  grows on the arc  $\cos u \leq -\beta$ , and the evaluation contour must be shrunk to  $|s| = r < \beta$ , rescaling the recovered coefficients by  $r^{-n}$ ; the conditioning trade-offs of the radius choice are analyzed in depth by Bornemann [4] (see also [1]).

### 3.1 The aliasing obstruction

**Theorem 3.4** (Integer-support aliasing). *Let  $y \in L^1_{\text{loc}}[0, \infty)$  and  $A \leq \infty$  be such that  $\widehat{y}(u) = \int_0^A y(\tau) e^{-\tau e^{iu}} d\tau$  converges for all  $u$ . Then  $\widehat{y}(u + 2\pi) = \widehat{y}(u)$  identically. Consequently, if  $\widehat{y}(u) = \int_{\mathbb{R}} \mathcal{B}(\alpha) e^{i\alpha u} d\alpha$  for a tempered distribution  $\mathcal{B}$ , then  $\text{supp } \mathcal{B} \subseteq \mathbb{Z}$ .*

*Proof.* The integrand depends on  $u$  only through  $e^{iu}$ . A  $2\pi$ -periodic tempered distribution has Fourier transform supported on the integer lattice (its Fourier series), which is the claim.  $\square$

**Corollary 3.5.** *No estimator that evaluates the (truncated) Laplace transform of measured data on the unit circle and inverts in  $u$  can produce mass at non-integer  $\alpha$ . In particular, fractional orders of a black-box system cannot be read off as non-integer spectral lines from such a construction, for any window  $A$ , sampling rate, or FFT length.*

The obstruction is a statement about the *readout*, not about the data: on  $\text{Re } s > 0$  the Laplace transform determines  $y$  uniquely, so the fractional information is present—it is the circle parametrization that projects it onto  $\mathbb{Z}$ . What survives the projection is a decay signature. For  $F(s) = s^{-1/2}$ , restricting the branch  $e^{-iu/2}$  to one period  $u \in (0, 2\pi)$  and expanding in a Fourier series,

$$c_m = \frac{1}{2\pi} \int_0^{2\pi} e^{-iu/2} e^{-imu} du = \frac{-i}{\pi(m + \frac{1}{2})}, \quad (8)$$

an integer comb decaying only like  $1/m$ —the fingerprint of a branch cut crossing the contour, as opposed to the geometric decay of an analytic  $H$ . Slow comb decay can thus *detect* non-integer content, but the order itself cannot be localized this way. When  $H$  is known as a formula, the lift to  $u \in \mathbb{R}$  (Section 2) is available and half-integer orders reappear at period  $4\pi$ ; for data, they do not.

## 4 The logarithmic-spiral estimator

The failure mechanism of the circle is that  $s = e^{iu}$  revisits the same point every  $2\pi$  while  $s^\alpha$ ,  $\alpha \notin \mathbb{Z}$ , does not. The fix is to sample where the contour never revisits: a logarithmic spiral

$$s_j = e^{(\sigma+i)u_j}, \quad u_j = u_0 + j \Delta u, \quad j = 0, \dots, N-1, \quad \sigma \neq 0, \quad (9)$$

single-valued along the contour. If the transform under study is a finite comb with distinct real orders  $\alpha_m$ , then its samples on the spiral take the form of a finite exponential sum in the index  $j$ :

$$F(s_j) = \sum_{m=1}^M c_m z_m^j, \quad z_m = e^{(\sigma+i)\Delta u \alpha_m}, \quad (10)$$

an exponential sum in the sample index  $j$ . Order identification is thereby converted into *line-spectrum estimation*, for which classical super-resolution tools exist; we use the matrix-pencil method of Hua and Sarkar [17] (Hankel data matrix, SVD, shifted singular-subspace pencil), which is exact for noiseless data with  $N \geq 2M$  and degrades gracefully with noise.

**Proposition 4.1** (Identifiability on the spiral). *For  $\sigma \neq 0$  and distinct real  $\alpha_m$ , the nodes  $z_m$  in (10) are distinct, and each  $\alpha_m$  is uniquely determined by  $z_m$ : indeed  $\ln |z_m| = \sigma \Delta u \alpha_m$  is single-valued, so  $\alpha_m = \ln |z_m| / (\sigma \Delta u)$  with no branch condition. If moreover  $|\Delta u \alpha_m| < \pi$ , the better-conditioned two-component formula  $\alpha_m = \text{Log } z_m / ((\sigma + i)\Delta u)$  applies, with Log the principal branch.*

*Proof.*  $z_m = z_{m'}$  requires  $(\sigma + i)\Delta u(\alpha_m - \alpha_{m'}) \in 2\pi i\mathbb{Z}$ ; the real part forces  $\sigma \Delta u(\alpha_m - \alpha_{m'}) = 0$ , hence  $\alpha_m = \alpha_{m'}$ . The moduli identity is immediate, and under  $|\Delta u \alpha_m| < \pi$  the exponent lies in the principal strip.  $\square$

The proposition isolates the degree of freedom the circle lacks: at  $\sigma = 0$  the moduli collapse to 1 and distinct orders can alias; within the family  $e^{(\sigma+i)u}$ , any  $\sigma \neq 0$  restores injectivity. (Sampling on the positive real axis,  $s = e^u$ , would also break the periodicity, at the cost of a classical real-node Prony problem; we return to a variant of it in Section 5.5.)

**Working from data.** For measured  $y$  the values  $F(s_j) = \hat{Y}_A(s_j)$  must converge, which requires  $\text{Re } s_j > 0$  when  $y$  decays slowly (fractional impulse responses decay algebraically; steady states not at all). We therefore restrict the spiral to an arc in the right half-plane,  $u \in [u_0, u_0 + (N-1)\Delta u] \subset (-\frac{\pi}{2}, \frac{\pi}{2})$ , on which the truncation error is bounded by  $e^{-A \min_j \text{Re } s_j}$  up to algebraic factors. Steady states need no special treatment here: a constant component of  $y$  is itself part of the order comb of  $\hat{Y}$  (mass  $y_\infty$  at  $\alpha = -1$ ) and is recovered as such, since the arc avoids the divergence region that plagued the circle. Two practical caveats: the model order  $M$  must be chosen (the singular-value profile of the pencil matrix gives a reliable indicator, cf. Section 5), and recovery of *continuous* order densities from a right-half-plane arc is a Laplace-type inversion and hence exponentially ill-posed—consistent with the conditional-stability results for distributed-order inverse problems [22]; the line-spectrum setting (finite combs) is the regime in which the method is exact.

## 5 Numerical experiments

All experiments are reproduced by `order_spectrum.py` (NumPy only); E5 by `cantor_pencil.py` and `deaf_channel.py`. Laplace transforms of sampled data are computed with composite Simpson quadrature.

### 5.1 E1: integer-order system from its step response

$H(s) = 1/(s^2 + 4s + 5)$ , poles  $-2 \pm i$  ( $\beta = 2$ ,  $|\text{poles}| = \sqrt{5}$ ), step response  $y(t) = \frac{1}{5} - \frac{1}{5}e^{-2t}(2 \sin t + \cos t)$  sampled with  $N_t = 65537$  points on  $[0, 12]$ ;  $N_u = 64$  circle points. The steady state is estimated as the mean of the last 2% of samples. Exact Taylor coefficients satisfy  $5a_n + 4a_{n-1} + a_{n-2} = 0$  for  $n \geq 2$ , with  $5a_0 = 1$  and  $5a_1 + 4a_0 = 0$ ; the corrected and naive estimates are compared in Table 1 below.



| $n$ | exact $a_n$ | corrected pipeline | naive pipeline        |
|-----|-------------|--------------------|-----------------------|
| 0   | +0.2        | +0.200000000005    | $2.5 \times 10^{-12}$ |
| 1   | -0.16       | -0.160000000057    | +2.24                 |
| 2   | +0.088      | +0.088000000323    | -14.31                |
| 3   | -0.0384     | -0.038400001209    | +57.56                |
| 4   | +0.01312    | +0.013120003331    | -172.8                |
| 5   | -0.002816   | -0.002816007179    | +414.7                |
| 6   | -0.0003712  | -0.000371187522    | -829.4                |

Table 1: E1. Corrected pipeline: max error  $1.2 \times 10^{-8}$  with estimated steady state ( $8.1 \times 10^{-9}$  if  $y_\infty$  is known). The naive pipeline is corrupted exactly as predicted by Proposition 3.2: at bin 1 the predicted excess is  $-y_\infty(-A) = +2.400$ ; the observed excess is +2.400. At bin 6,  $b_5 = -0.2 \cdot 12^6/6! = -829.44$  matches all printed digits.

## 5.2 E2: the aliasing signature of a branch cut

Sampling the branch  $F(e^{iu}) = e^{-iu/2}$  of  $s^{-1/2}$  at  $N_u = 64$  points of one period and FFT-ing yields bins matching the closed form (8):  $|\text{bin}_0| = 0.6367$  vs.  $1/(\pi \cdot \frac{1}{2}) = 0.6366$ ;  $|\text{bin}_3| = 0.0914$  vs. 0.0909;  $|\text{bin}_{10}| = 0.0317$  vs. 0.0303 (the residuals combine the  $m + \ell N_u$  aliasing sums with a  $1/N_u = 0.0156$  offset from sampling the branch-cut discontinuity at  $u = 0$ ; the latter dominates in the high bins). Energy spreads over *all* integer bins with  $1/(m + \frac{1}{2})$  decay; nothing appears at  $\alpha = -\frac{1}{2}$ , illustrating Theorem 3.4.

## 5.3 E3: spiral estimator, exact fractional samples

$H(s) = s^{-1/2} + 0.5s^{1.3}$  sampled on the spiral  $\sigma = 0.2$ ,  $\Delta u = 0.3$ . With  $N = 8$  noise-free samples the matrix pencil returns  $\alpha = (-0.500000000000, +1.300000000000)$ , weights (1.0, 0.5), order-position error  $5 \times 10^{-16}$ . With  $N = 64$  and relative sample noise  $10^{-8}$  (resp.  $10^{-4}$ ) the order errors are  $4.7 \times 10^{-9}$  (resp.  $4.7 \times 10^{-5}$ ): errors scale linearly with the noise floor.

## 5.4 E4: fractional identification from raw time samples

$y(t) = 1/\sqrt{\pi t} + 2$ , whose transform is  $\hat{Y}(s) = s^{-1/2} + 2s^{-1}$ : a two-line order comb with one fractional line. The substitution  $t = v^2$  removes the integrable singularity ( $\int_0^A y e^{-st} dt = \int_0^{\sqrt{A}} (2/\sqrt{\pi} + 4v) e^{-sv^2} dv$ ), sampled with  $N_v = 32769$  points,  $A = 120$ . The spiral arc  $\sigma = 0.15$ ,  $u \in [-1.35, 1.35]$ ,  $N = 64$  lies in  $\text{Re } s \geq 0.17$ ; the recovered orders and weights are reported in Table 2 below.

| noise (rel., on $y$ ) | $\hat{\alpha}_1$ | $\hat{\alpha}_2$ | $\hat{c}_1$ | $\hat{c}_2$ |
|-----------------------|------------------|------------------|-------------|-------------|
| 0                     | -1.00000000      | -0.50000000      | 2.00000     | 1.00000     |
| $10^{-4}$             | -0.99997350      | -0.49995259      | 2.00021     | 0.99979     |

Table 2: E4. True  $(\alpha, c) = (-1, 2), (-\frac{1}{2}, 1)$ . Noise-free order error  $1.8 \times 10^{-9}$ ; at  $10^{-4}$  noise,  $4.7 \times 10^{-5}$ . The pencil singular values (97.1, 0.89,  $1.6 \times 10^{-9}, \dots$  noise-free; third value rising to  $7.8 \times 10^{-6}$  under noise) clearly indicate the model order  $M = 2$ .

Experiment E4 performs, from time-domain samples alone, precisely the task that Corollary 3.5 proves impossible for the unit-circle readout: localization of a non-integer order. The steady-state-like component ( $2/s$ ) is recovered as the  $\alpha = -1$  line rather than subtracted, as promised in Section 4.

## 5.5 E5: a calibration standard from fractal geometry—complex dimensions of the Cantor string

Experiments E1–E4 validate the estimator against synthetic signals drawn from the very model class the method assumes; the ground truth is planted by the experimenter. A sharper test requires systems whose exponential-sum structure is a *theorem* rather than a construction, with parameters known through an entirely independent computational route, and—ideally—available in families where a dichotomy predicted by theory (uniform versus non-uniform damping) is *certified* rather than conjectural. The theory of fractal strings supplies exactly this: a bounded open set  $\mathcal{L} = \{\ell_j\} \subset \mathbb{R}$  has geometric zeta function  $\zeta_{\mathcal{L}}(s) = \sum_j \ell_j^s$ —an instance of the representation (1) with order comb at the log-lengths—whose poles, the *complex dimensions*, govern the string’s geometric and spectral oscillations [20, 21]. Self-similar (“lattice”) strings have all complex dimensions on one vertical line, a theorem rather than a hypothesis; this is the calibration standard used below.

**The system.** The Cantor string (complement of the middle-thirds Cantor set in  $[0, 1]$ ) has lengths  $3^{-n}$  with multiplicities  $2^{n-1}$ ,  $\zeta_{\mathcal{L}}(s) = 3^{-s}/(1 - 2 \cdot 3^{-s})$ , and complex dimensions known in closed form:

$$\begin{aligned} \omega_k &= D + ikp, & D &= \frac{\log 2}{\log 3} = 0.6309\,2975\dots, \\ p &= \frac{2\pi}{\log 3} = 5.7192\,0173\dots, & k &\in \mathbb{Z}, \end{aligned} \tag{11}$$

all on the single vertical line  $\operatorname{Re} s = D$ : the *lattice case*. The frequency (spectral) counting function of the associated Dirichlet Laplacian satisfies the spectral factorization  $\zeta_{\nu}(s) = \zeta_{\mathcal{L}}(s) \zeta(s)$ , so each geometric oscillation reaches the spectrum multiplied by the channel gain  $\zeta(\omega_k)$ ; the Lapidus–Maier inverse spectral theorem [19] identifies zeros of  $\zeta$  with null frequencies of this geometry-to-spectrum channel, with the critical line as the only possibly-deaf band.

**Method.** In the logarithmic variable  $\lambda = \log x$ , Poisson summation gives the Cantor string’s geometric measure *exactly* as a mode expansion,

$$\sum_{n \geq 1} 2^{n-1} \delta(\lambda - n \log 3) = \frac{1}{2 \log 3} \sum_{k \in \mathbb{Z}} e^{\omega_k \lambda}$$

(no error term—in contrast with arithmetic combs such as the prime-power comb, where the analogous expansion carries an explicit-formula error term). Gaussian smoothing by  $\varphi_h$  multiplies mode  $k$  by  $e^{\omega_k^2 h^2 / 2}$ , truncating the tower to an effective model order exactly as in Section 4; sampling at  $\lambda_j = \lambda_0 + j\Delta$  produces the exponential-sum model (10) with nodes  $z_k = e^{\omega_k \Delta}$ , and the matrix pencil returns  $\operatorname{Re} \hat{\omega} = \ln |z| / \Delta$  (the damping/Minkowski-dimension line) and  $\operatorname{Im} \hat{\omega} = \arg z / \Delta$  (oscillatory dimensions), by the argument of Proposition 4.1 (with the roles of contour and orders interchanged: a real logarithmic grid resolving complex orders). For the spectral run, the smoothed spectral measure has atoms at  $\lambda = \log(m \cdot 3^n)$ ; its smooth part is the Weyl term  $e^\lambda$  (total length 1, the string’s steady state, subtracted exactly as in Corollary 3.3) plus modes weighted by  $\zeta(\omega_k)/(2 \log 3)$ . Parameters throughout:  $h = 0.10$ ,  $\Delta = 0.07$ ,  $N = 64$ ,  $\lambda_0 = 5.0$ , pencil parameter  $N/2$ , energy-threshold retention, a priori filter  $\operatorname{Re} \hat{\omega} \in (0.2, 1.2)$ .

**F1: complex dimensions from geometry.** Input: the lengths only.

Every real part equals  $D$  to the printed six decimals: the lattice theorem, measured by the instrument. A ninth node ( $k = 8$ ) emerges degraded at the smoothing floor  $e^{-k^2 p^2 h^2 / 2}$ , exactly where the weight ladder predicts the retained set to end.



| $\text{Re } \hat{\omega}$ | $\text{Im } \hat{\omega}$ | exact $\text{Im} = kp$ | $k$ |
|---------------------------|---------------------------|------------------------|-----|
| 0.630930                  | 0.000000                  | 0.000000               | 0   |
| 0.630930                  | 5.719202                  | 5.719202               | 1   |
| 0.630930                  | 11.438403                 | 11.438403              | 2   |
| 0.630930                  | 17.157605                 | 17.157605              | 3   |
| 0.630930                  | 22.876807                 | 22.876807              | 4   |
| 0.630930                  | 28.596009                 | 28.596009              | 5   |
| 0.630930                  | 34.315211                 | 34.315210              | 6   |
| 0.630928                  | 40.034412                 | 40.034412              | 7   |

Table 3: E5/F1: complex dimensions of the Cantor string recovered from geometric data alone. Every real part equals  $D = \log 2 / \log 3$  to the printed six decimals: the lattice theorem measured by the instrument—a uniformly damped order comb, here certified truth rather than conjecture.

**F2: the same dimensions from the sound, with  $\zeta$  as the measured gain.** Input: the spectral counting data, Weyl term subtracted. The pencil recovers the same tower ( $\text{Re } \hat{\omega} = 0.630930$  for  $k \leq 5$ , drifting to 0.630937 at the floor), and the amplitude ratio of each spectral mode to its geometric counterpart matches the channel gain independently computed from  $\zeta$  to four decimals in every one of eight modes (e.g.  $k = 0$ : measured 2.1598 vs.  $|\zeta(D)| = 2.1598$ ;  $k = 5$ : measured 2.2269 vs. 2.2269): an *acoustic measurement of  $|\zeta|$  along the vertical line  $\text{Re } s = D$* , obtained entirely from integer counting data, realizing the spectral factorization  $\zeta_\nu = \zeta_{\mathcal{L}} \cdot \zeta$  as a live, measured transfer function.

**F3: the deaf channel at the critical line.** To exhibit the Lapidus–Maier obstruction directly, a generalized string with geometric counting function  $N_{\mathcal{L}}(y) = y^{1/2} + \varepsilon \text{Re}(y^{1/2+i\gamma})$ ,  $\varepsilon = 0.5$  (geometry oscillating at complex dimension  $\frac{1}{2} + i\gamma$ ), gives, after computing  $N_\nu$  by direct summation (no  $\zeta$  anywhere in the forward model) and subtracting the Weyl term: a control frequency  $\gamma = 10.0$  (not a zero) survives with measured amplitude 0.7747 against a predicted  $\varepsilon|\zeta(\frac{1}{2} + 10i)| = 0.7746$ ; the geometric oscillation at  $\gamma = 14.134725$  (the first zeta zero), of the same input amplitude 0.5, survives with measured amplitude **0.0007** against a predicted 0.0000. A fractal string can hide information from its own sound precisely at the zeta zeros; the estimator’s blind spots, read off as measured channel gains, *are* the zeros—the inverse-spectral content of the Riemann Hypothesis made visible as an instrument reading, with no claim of bearing on the Hypothesis itself: the zeros are the kernel of a forward map, and information annihilated by a forward map is unrecoverable by any estimator.

**Why this experiment matters for the estimator.** F1–F3 supply three validation claims that E1–E4 cannot. (i) *End-to-end correctness against an external, independently-computed standard*, in a regime (F1) where the target values are proven rather than conjectured. (ii) *The complex-order regime*: the pencil and the modulus formula of Proposition 4.1 resolve complex orders, and F2 shows the recovered *amplitudes*, not just positions, carry independently verifiable physical content (the values of  $\zeta$  itself). (iii) A *predicted failure mode*, deliberately engineered (F3): the estimator’s channel gain vanishes exactly where theory says it must, demonstrating that the method’s blind spots are structural, not accidental, and are located precisely where the theory of Section 3–4 predicts.

## 6 Related work

**Distributed-order systems and order distributions.** Equation (1) is the transfer function of distributed-order systems, originating with Caputo [5, 6, 7] and developed as an “order domain” by Bagley and Torvik [2, 3]; Lorenzo and Hartley formalized variable- and distributed-order operators [23] and the Laplace-domain operator classes that order distributions generate [15], and the monograph [18] treats stability and simulation of the class; see [8] for a survey. The identification of order distributions was posed and attacked by Hartley and Lorenzo [13, 14]: they already observe the structural identity behind (2) (the order integral as a Laplace transform in  $-\log s$ , their  $r$ -Laplace transform, Eq. (18) of [13]) and represent integer systems as impulsive order distributions; their estimator, however, discretizes the order axis and solves least squares against frequency-response data, as do its successors [26, 9]. Optimization- and instrumental-variable estimation of discrete fractional orders is developed in [27, 28]; log-periodogram regression estimates long-memory orders in time series [11]. The present paper differs in estimator class: closed-form/spectral rather than fitted, with the obstruction theorem delimiting where the spectral route is possible.

**Cauchy-integral coefficient extraction.** Computing Taylor–Laurent coefficients by trapezoidal Cauchy integrals on circles is classical [25, 16, 10], with definitive accuracy and radius-selection analysis in [4] and a modern roots-of-unity framework in [1]. Corollary 3.3 identifies our corrected integer-order pipeline with this method applied to a transform computed from data; the adjacent inverse direction (Weeks-type inversion, unit-circle FFTs of Laplace-domain functions) is surveyed in [29], and fully numerical *forward* Laplace transforms of sampled data are treated in [30]. The information content of the integer bins is that of classical time moments [12].

**What is new here.** To our knowledge: the exact bias formula (6) for step-response truncation on circle contours; the aliasing theorem (Theorem 3.4) with the branch-cut decay signature (8) as the data-side limit of the unit-circle program; the identifiability statement of Proposition 4.1; and the composition *data*  $\rightarrow$  *truncated Laplace transform on a right-half-plane spiral arc*  $\rightarrow$  *matrix-pencil line spectrum*  $\rightarrow$  *orders and weights*, validated from raw time samples; and the use of fractal-string complex dimensions as a calibration benchmark for order-distribution estimators (E5), including an acoustic measurement of the Riemann zeta function along a vertical line and a demonstration of the estimator’s channel gain vanishing at a zeta zero. Each ingredient separately is standard.

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## Conflict of Interest

The author declares no conflict of interest.

## Data Availability Statement

No experimental or proprietary data were used in this work; all numerical experiments (Section 5) use synthetic signals generated in closed form by the accompanying code, including E5, whose geometric and spectral counting data are generated in closed form from the Cantor

string’s defining lengths within the code itself. The complete source code implementing every experiment, together with the exact numerical output validating it, is openly available at <https://drive.google.com/file/d/1eselh0HetF4mnDnc4iptheXgnBU57oXE/view> (see Appendix B for details).

## Declaration of AI-Assisted Technologies

During preparation of this work the author used a large language model (LLM) to verify derivations, search prior literature, help derive Propositions 3.1 and 3.2 and Theorem 3.4, design and implement the accompanying numerical code. After using this tool the author reviewed and edited all content as needed and takes full responsibility for the content of this publication.

## A Proofs

*Proof of Proposition 3.1.* Expanding  $e^{-\tau e^{iu}} = \sum_{j \geq 0} \frac{(-\tau)^j}{j!} e^{iju}$  (absolutely convergent, uniformly on compacts) and integrating over  $[0, A]$  termwise gives  $\widehat{y}(u) = \sum_{j \geq 0} \mu_j e^{iju}$  with  $\mu_j = \frac{(-1)^j}{j!} \int_0^A \tau^j y(\tau) d\tau$ ; the DFT of a uniform sampling of  $\sum_j \mu_j e^{iju}$  aliases  $j$  modulo  $N$  into the stated sum.  $\square$

*Proof of Proposition 3.2.* Write  $\int_0^A = \int_0^\infty - \int_A^\infty$  on both components of  $y = y_\infty + g$ . The constant part gives  $y_\infty(1 - e^{-sA})/s$ , whose expansion

$$\begin{aligned} \frac{y_\infty(1 - e^{-sA})}{s} &= \frac{y_\infty}{s} \left( 1 - \sum_{k \geq 0} \frac{(-sA)^k}{k!} \right) = -y_\infty \sum_{k \geq 1} \frac{(-A)^k s^{k-1}}{k!} \\ &= -y_\infty \sum_{m \geq 0} \frac{(-A)^{m+1}}{(m+1)!} s^m \end{aligned}$$

is entire: the pole  $y_\infty/s$  cancels exactly (which is why no mass appears at  $\alpha = -1$ ) and the coefficient of  $s^m$  is (6). The transient part contributes  $G(s)$  minus its tail, bounded as stated since  $|e^{-s\tau}| = e^{-\tau \cos u}$  on  $|s| = 1$ .  $\square$

*Note.* The Poisson-summation and Gaussian-smoothing identities used in E5 follow the standard explicit-formula machinery for self-similar fractal strings [21]: the geometric measure of the Cantor string is log-periodic after renormalization by  $e^{-D\lambda}$ , so the Gaussian–exponential integrals evaluate exactly to  $\int \varphi_h(u) e^{-\omega u} du = e^{\omega^2 h^2/2}$  for each complex dimension  $\omega = \omega_k$ , with no further error term—in contrast with arithmetic combs such as the prime-power comb.

## B Reproducibility

`order_spectrum.py` (NumPy) implements `circle_taylor_coeffs` (Corollary 3.3), `naive_circle_coeffs` (for Table 1’s bias column), `matrix_pencil` [17], and `spiral_order_estimate` (Section 4); running the file reproduces E1–E4 verbatim. `cantor_pencil.py` (NumPy + mpmath, the latter only for the independent reference  $|\zeta|$  values used to check F2; no zeta values enter the forward data) reproduces F1 and F2 and Table 3; `deaf_channel.py` reproduces F3. Code available at <https://drive.google.com/file/d/1eselh0HetF4mnDnc4iptheXgnBU57oXE/view> (`order_spectrum.py`, E1–E4), [https://drive.google.com/file/d/1AMGYCC5RAgxeH3odohpvt\\_nBpiDiR6zQ/view](https://drive.google.com/file/d/1AMGYCC5RAgxeH3odohpvt_nBpiDiR6zQ/view) (`cantor_pencil.py`, F1–F2), and <https://drive.google.com/file/d/1Giu4cQ-A-zB85gdzpZ1xmXw0NMUc607U/view> (`deaf_channel.py`, F3).

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